



Portfolio website

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"An enthusiastic Data Science undergraduate skilled in full-stack development and data-driven problem solving, who is passionate about learning new things and applying them in practical scenarios to make an advancement in people's lives."

PROJECTS

Kidney Disease Classification

Python, CNN, MLflow, DVC, Docker, Flask

- Built an end-to-end deep learning pipeline for kidney disease classification, covering model training through deployment to support data-driven healthcare analysis.
- Integrated experiment tracking and pipeline versioning with MLflow and DVC to improve reliability, transparency, and performance monitoring of predictions.
- Containerized and deployed the model-serving API using Docker and Flask.

Data-Efficient Vision Transformers for Small Image Datasets

Python, PyTorch, timm, CIFAR-10

- Designed a controlled experiment comparing a Vision Transformer trained from scratch, a ResNet-18 CNN, and an ImageNet-pretrained Vision Transformer on CIFAR-10 to study why Transformers need more data than CNNs to generalize well.
- Built an end-to-end pipeline (data loading, training with AdamW + cosine LR scheduling, evaluation, and visualization) with a CLI runner supporting configurable data subsets and epoch counts.
- Validated the hypothesis on an initial smoke test (2% data, 1 epoch): pretrained ViT reached 74.7% accuracy (0.728 macro F1) versus 17.9% for the CNN and 17.2% for the from-scratch ViT, confirming pretraining closes the data-efficiency gap.
- Added Grad-CAM-style attention map visualizations showing the pretrained ViT attends to the actual object while the from-scratch ViT's attention remains diffuse and unfocused.

Content Crew - Multi-Agent AI Content Platform

CrewAI, Groq (Llama 3.1), FastAPI, PostgreSQL (Neon), Google OAuth, Docker

- Built a three-agent sequential pipeline (Researcher, Writer, Editor) with CrewAI, where the Researcher agent performs live DuckDuckGo web search and each agent step is powered by Groq's Llama 3.1 8B Instant model.
- Implemented Google OAuth sign-in with per-user, session-scoped article storage in PostgreSQL (Neon), plus Markdown/PDF/Word export and shareable read-only public links.
- Added automatic exponential-backoff retry for failed LLM calls and a batch-mode CLI for generating multiple articles from a topic list; containerized with Docker and deployed on Hugging Face Spaces.

Multi-label Chest X-ray Disease Classifier

Python, PyTorch, ResNet50, EfficientNet-B0, Grad-CAM, FastAPI, React, Docker

- Built a multi-label chest X-ray disease classifier on the NIH ChestX-ray14 dataset, training a custom CNN baseline and comparing it against ResNet50 and EfficientNet-B0 transfer-learning models.
- Handled severe class imbalance with weighted loss and evaluated performance using per-class AUC-ROC and F1 metrics, improving ResNet50 macro F1 from 0.308 to 0.318 through threshold tuning.
- Added Grad-CAM visualizations for model interpretability and deployed a full-stack demo (FastAPI backend, React/Vite/Tailwind frontend) on Hugging Face Spaces via Docker.

AVJNS -Anonymous Victim-Centered Justice Navigation System

Forked - Python, FastAPI, Groq, Supabase (pgvector), Next.js

- Studied and ran a supervised multi-agent legal-guidance system for Sri Lanka, exploring its Supervisor-orchestrated architecture across Safety, Triage, Risk, Intake, Retrieval, Guidance, Critic, Crisis, and Navigation sub-agents over a shared case-state graph.
- Reviewed its implementation of Corrective RAG over a Supabase pgvector knowledge base, and a Reflexion-style Guardrail Critic that reviews every draft against hard safety rules before it reaches the user.

UniMate – University Lifestyle Platform

React, Node.js, Express, MongoDB

- Designed and developed a full-stack university lifestyle platform addressing student needs in accommodation, services, and academic support.
- Built marketplace access, study resources, and service discovery features with CRUD functionality and role-based access control for secure, efficient data management.

Healthcare Experience & Quality Analysis

Python, pandas, scikit-learn, SciPy, Seaborn

- Conducted a statistical analysis of patient survey data to test whether comfort-related healthcare factors (waiting time, doctor quality, diagnosis accuracy, facilities) significantly affect overall patient satisfaction.
- Applied Spearman correlation, Kruskal-Wallis testing, and predictive modeling (logistic regression, random forest) to validate the hypothesis at a 95% confidence level.

Hotel Data Warehouse – SQL Server Analysis Services Cube

SQL Server, SSAS, OLAP, Star Schema

- Built an OLAP cube for hotel booking analytics using a star-schema data warehouse design to support business intelligence

n8n Email Triage Bot

n8n, Groq (Llama 3.3 70B), Gmail API, Slack API, Notion API, Google Sheets API, Docker

- Built a self-hosted n8n workflow that monitors a Gmail inbox and classifies each incoming support email into Urgent, Billing, or General using Llama 3.3 70B via Groq's API, routing tickets to Slack alerts, Notion ticket pages, or an automated FAQ reply.
- Designed a loop-guard node to prevent the bot from re-classifying and re-replying to its own auto-generated responses, and logged every ticket with a confirmation email and a Google Sheets audit row.

Online Vaccination Portal

HTML, CSS, JavaScript, PHP, MySQL

- Contributed to an online vaccination coordination portal covering appointment handling, staff roles, and inventory updates, with data tracking and basic analytics for visibility into vaccination progress.

Online Help Desk System

HTML, CSS, JavaScript, PHP

- Contributed to a structured ticketing platform for managing and resolving student queries, supporting organized data handling for issue tracking and response efficiency.

EDUCATION

BSc (Hons) in Information Technology, Specializing in Data Science

GPA: 3.76

SLIIT - Sri Lanka Institute of Information Technology

G.C.E. Advanced Level (A/L) – Technology Stream

INTERESTS

Full-stack developer | Artificial Intelligence | Generative AI | Deep Learning | Machine Learning |Data Science | Software Engineering

SKILLS

Programming Languages: Python, JavaScript, TypeScript, Java

Web & Backend Development: Node.js, Express.js, Django, FastAPI, REST API Design

Machine Learning & Data Science: CNN, PyTorch, ResNet50/EfficientNet, MLflow, DVC, scikit-learn, pandas

Mobile Development: Android (Kotlin), MVVM Architecture

Databases: MongoDB, MySQL, PostgreSQL, Firebase

REFERENCES

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